Huiying (Helen) Li

Contact Information

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Industry Experience

AI Research Scientist, Meta *Apr. 2026 - present*

- Conduct research on personalized AGI, building AI systems that understand each user's preferences, context, and goals to deliver tailored and genuinely helpful experiences.
- Leverage expertise in large-scale recommendation and user modeling to advance personalization capabilities of foundation models.

Principal Machine Learning Scientist, TikTok *Apr. 2023 - Apr. 2026*

- Develop and optimize large-scale recommendation models to enhance the personalized content discovery experience for users on TikTok ForU page.
- Balance user content consumption with business objectives, optimizing user experience while aligning with organizational goals.

PhD Software Engineer Intern, Meta *Jun. 2022 - Sept. 2022*

- Implemented my work "Blacklight: Scalable Defense for Neural Networks against Query-Based Black-Box Attacks." (USENIX Security'22) with the AI Security Team @ Meta.

Research Intern, Microsoft Research *Jun. 2020 - Sept. 2020*

- Worked with the Security + AI RIP group @Microsoft Research on measuring the utility of defenses against adversarial ML attacks.

Education

The University of Chicago, Chicago, IL, United States *Sept. 2017 - Mar. 2023*
Ph.D. in Computer Science *GPA:* **4.0/4.0**

The University of Chicago, Chicago, IL, United States *Sept. 2017 - Mar. 2020*
M.S. in Computer Science

Fudan University, Shanghai, China *Sept. 2013 - Jun. 2017*
B.S. in Computer Science and Technology

Awards

- Siebel Scholarship (2021)
- CHI Honorable Mention Award (2020)
- Facebook Fellowship (2020)
- Two Sigma Fellowship Finalist (2020)
- National Scholarship, China (**Top 0.2%** in China) (2015)
- Outstanding Student, Fudan University (2014)

Academic Experience

Research Assistant, SAND Lab, University of Chicago *Sept. 2017 - Mar. 2023*

Supervised by Prof. Ben Y. Zhao and Prof. Heather Zheng

Teaching Assistant, University of Chicago
CMSC 23360 Advanced Networks *Spring 2021*
CMSC 23400 Mobile Computing *Winter 2018*
MPCS 52011 Introduction to Computer Systems *Autumn 2017*

Reviewer, IEEE International Conference on Acoustics, Speech and Signal Processing,

2023

Reviewer, ICLR Backdoor Attacks and Defenses in Machine Learning Workshop, 2023

Shadow Program Committee, IEEE Symposium on Security and Privacy, 2021

Research

SAND Lab, University of Chicago *Sept. 2017 - Mar. 2023*
Ph.D. Student Supervised by Prof. Ben Y. Zhao and Prof. Heather Zheng

- **ML Security and Robustness**
  - Attacks and defenses for DNN backdoor attacks.
  - Defenses for adversarial attacks against DNNs.
- **Human Privacy Protection**
  - Image "Cloaking" for human facial privacy.
  - Wearable jammer against commercial microphones.

Publication

Huiying Li, Arjun Nitin Bhagoji, Ben Y Zhao, Haitao Zheng. "Can Backdoor Attacks Survive Time-Varying Models?" arXiv preprint arXiv:2206.04677 (2022).

Huiying Li, Shawn Shan, Emily Wenger, Jiayun Zhang, Haitao Zheng, Ben Y. Zhao. "Blacklight: Scalable Defense for Neural Networks against Query-Based Black-Box Attacks." In Proceedings of *The 31th USENIX Security Symposium.* Boston, MA, Aug. 2022. **USENIX Security'22**

Huiying Li, Emily Wenger, Shawn Shan, Ben Y. Zhao, Haitao Zheng. "Piracy Resistant Watermarks for Deep Neural Networks." arXiv preprint arXiv:1910.01226 (2020).

Shawn Shan, Emily Wenger, Jiayun Zhang, **Huiying Li**, Haitao Zheng, Ben Y. Zhao. "Fawkes: Protecting Privacy against Unauthorized Deep Learning Models." In Proceedings of *The 29th USENIX Security Symposium.* Boston, MA, Aug. 2020. **USENIX Security'20**

Yuxin Chen*, **Huiying Li***, Shan-Yuan Teng*, Steven Nagels, Zhijing Li, Pedro Lopes, Ben Y. Zhao, and Haitao Zheng. "Wearable Microphone Jamming." In Proceedings of *The CHI Conference on Human Factors in Computing Systems.* Honolulu, HI, Apr. 2020. **ACM CHI'20** (**Honorable Mention Award**)
** denotes equal contribution*

Yuanshun Yao, **Huiying Li**, Haitao Zheng, and Ben Y. Zhao. "Latent Backdoor Attacks on Deep Neural Networks." In Proceedings of *The 26th ACM Conference on Computer and Communication Security.* London, UK, Nov. 2019. **ACM CCS'19**

Bolun Wang, Yuanshun Yao, Shawn Shan, **Huiying Li**, Bimal Viswanath, Haitao Zheng, and Ben Y. Zhao. "Neural Cleanse: Identifying and Mitigating Backdoor Attacks in Neural Networks." In Proceedings of *The 40th IEEE Symposium on Security and Privacy.* San Francisco, CA, May 2019. **IEEE S&P'19**

Talks

Conference talk at USENIX Security 2022
Blacklight: Scalable Defense for Neural Networks against Query-Based Black-Box Attacks

Conference talk at ACM CCS 2019
Latent Backdoor Attacks on Deep Neural Networks

Invited talk at EE380 Stanford
Persistent and Unforgeable Watermarks for Deep Neural Networks